

Predicting stop codon readthrough from mRNA sequence context using neural network models

A THESIS

SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENTS FOR THE DEGREE OF
Master of Science (Research)

by

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CERTIFICATE

This is to certify that the thesis titled "Predicting stop codon readthrough from mRNA sequence context using neural network models" is the outcome of original work carried out by Aditi Arun for the degree of Master of Science (Research) with a Major in Biology at the Indian Institute of Science, Bangalore, India.



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DECLARATION

The work reported in this thesis is original and was carried out by me during my tenure as a master's student at the Indian Institute of Science, Bangalore. This thesis has not formed the basis for the award of any degree, diploma, associateship, membership, or similar title of any university or institution.

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Acknowledgements

I'd like to thank my PI Dr. Sandeep Eswarappa, for his support and understanding throughout the thesis project. Also, a big thank you to Prof. Nagasuma Chandra for consulting on this project and providing key pointers and guidance on computational modelling. Last but not least, I'm very grateful to my parents and friends Sourabh, Oviya, Divija, Diya and Rahul for all of the fun and memories that have enriched my college life.

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Chapter 1

Introduction

Although the genetic code is largely considered universal and well defined, various anomalies exist in practice that challenge this assertion. One such anomaly is the biological phenomenon of stop codon readthrough or translational readthrough. During this process, a ribosome defies the conventional genetic code and continues translation past the stop codon into the 3' untranslated region. Thus, instead of terminating translation at the stop codon, the ribosome decodes the stop codon as an amino acid and continues translating until it encounters the next in-frame stop codon, leading to the production of a protein that is elongated at the C-terminus.

While the exact molecular mechanisms underlying translational readthrough are yet to be understood, it is a tightly regulated process and is known to be influenced by various factors such as the identity of the stop codon (UGA, UAG, or UAA), the sequence context surrounding the stop codon, and the availability of release factors and near-cognate tRNA.

1.1. Significance of translational readthrough

The ability of the ribosome to bypass a stop codon is not merely an evolutionary accident; it serves as a critical regulatory layer in both healthy physiology and the pathology of genetic diseases. The significance of studying this phenomenon can be viewed through two primary lenses: the functional expansion of the natural proteome and the therapeutic potential for correcting genetic mutations [7].

1.1.1 Functional diversity through natural readthrough

During translation of the endogenous Natural Termination Codons (NTCs), translational readthrough acts as an avenue for proteome expansion. By translating beyond the standard stop signal, a single mRNA transcript can generate two distinct protein isoforms from the same gene—the canonical protein and a C-terminally extended version.

These extensions often carry unique functional motifs that fundamentally alter the protein's behaviour:

- **Regulation of Protein Localisation:** Readthrough can unmask hidden "ZIP codes" within the 3' UTR. For example, the enzymes LDHB and MDH1 generate isoforms with a Peroxisomal Targeting Sequence (PTS) in their extended tail, redirecting these metabolic proteins from the cytosol to the peroxisome to assist in redox homeostasis.
- **Regulation of Protein Expression:** In genes like MTCH2, the double-readthrough product is highly unstable and prone to proteasomal degradation. This allows the cell to rapidly tune total protein levels by simply adjusting readthrough efficiency. In other cases, such as AMD1, readthrough causes ribosomes to stall, creating a "traffic jam" that inhibits further translation of the main protein.

- Modulation of Protein Function:** Protein extension by readthrough can also toggle its primary activity. The readthrough isoform of VEGFA (known as VEGF-Ax) cannot bind its co-receptor, Neuropilin-1. Consequently, while the canonical protein promotes blood vessel growth (angiogenesis), the readthrough version acts as an inhibitor, providing a natural check on tumour growth.

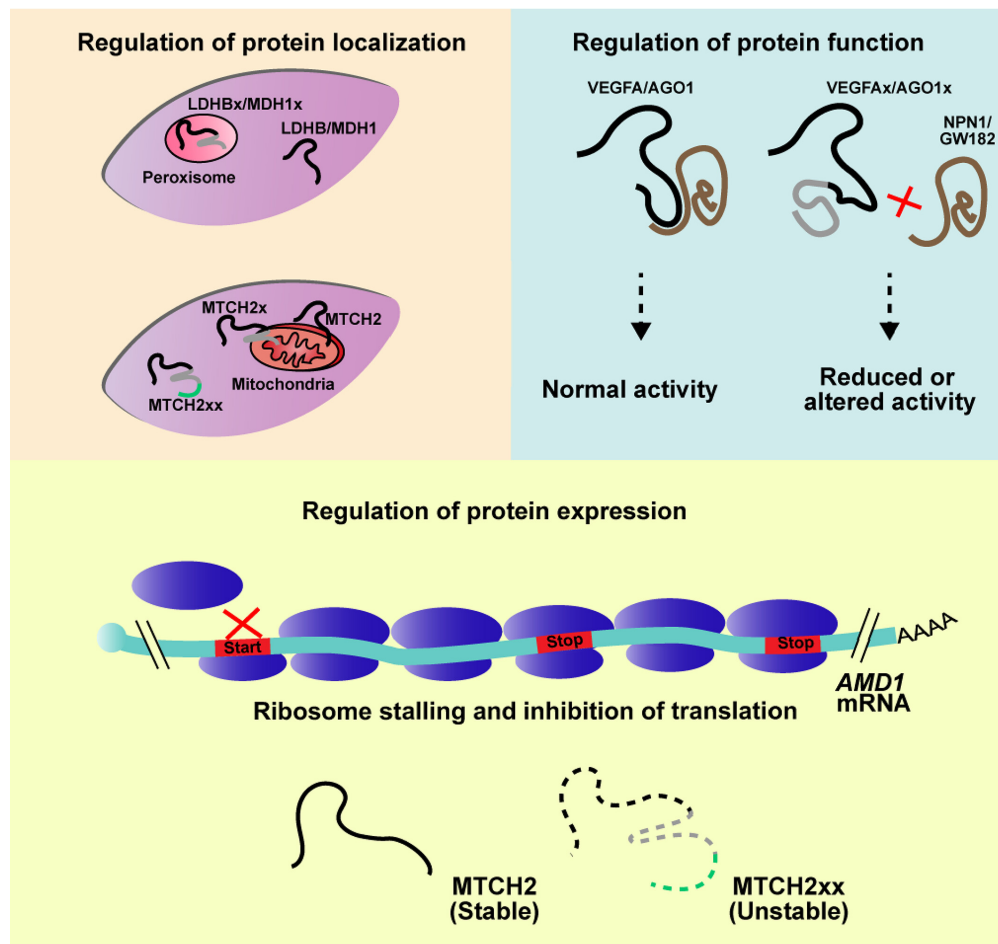


Figure 1.1: Readthrough significance across different conditions.

1.1.2 Premature termination codons (PTCs) and genetic disease

While readthrough at NTCs acts as a regulated tool for diversity, the introduction of Premature Termination Codons (PTCs) represents a severe pathological challenge. PTCs arise from nonsense mutations that place a stop signal in the middle of a coding sequence, halting translation before the protein is complete.

PTCs are the underlying cause of approximately 11% of all inherited genetic disorders, including cystic fibrosis, Duchenne muscular dystrophy, and various hereditary cancers. These mutations are devastating because they lead to truncated, non-functional proteins and often trigger the degradation of the mRNA itself through the Nonsense-Mediated Decay (NMD) pathway.

1.1.3 Therapeutic readthrough and the problem of specificity

The inherent "leakiness" of the translation machinery suggests a promising therapeutic strategy: if we can artificially induce readthrough at a disease-causing PTC, we can restore the production of full-length, functional proteins. This approach, known as "nonsense suppression therapy," aims to treat the symptoms of a detrimental mutation by exploiting the ribosome's ability to occasionally bypass stop signals.

To achieve this, researchers utilise specific classes of small molecules, such as certain antibiotics and synthetic compounds, designed to promote translational readthrough. These drugs interfere with the ribosome's decoding process, making it more likely to mistake a stop codon for a sense codon, and allowing a "near-cognate" tRNA to insert an amino acid at the mutation site and continue translation to the end of the transcript to produce a functional protein.

However, the clinical utility of these drugs is severely hindered by a fundamental lack of specificity. Because aminoglycosides target the ribosome itself rather than a specific mRNA sequence, they act as non-specific catalysts for readthrough across the entire transcriptome.

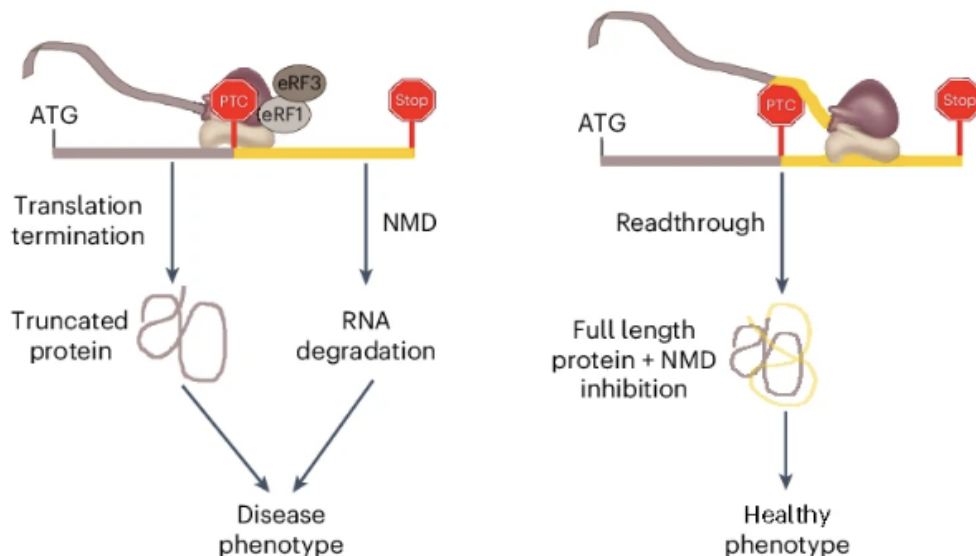


Figure 1.2: Readthrough significance across different conditions.

While the goal is to target "bad" PTCs to restore a missing protein, the drug simultaneously promotes readthrough in "good" Natural Termination Codons (NTCs) in thousands of healthy genes. This can lead to a state of global readthrough, where the cell begins to produce an array of C-terminally extended protein isoforms that would not exist under normal physiological conditions.

The consequences of this unchecked readthrough can be physiologically devastating. When NTCs are bypassed at high frequencies, the resulting protein extensions can disrupt delicate cellular balances. For example, excessive readthrough of MPZ (Myelin Protein Zero) in the peripheral nervous system or AQP4 (Aquaporin-4) in the brain can create autoantigens—extended proteins that the immune system no longer recognises as "self"—potentially triggering autoimmune responses. Furthermore, these extensions can lead to protein misfolding or the shielding of vital functional domains, contributing to the severe nephrotoxicity and ototoxicity frequently observed in patients undergoing long-term aminoglycoside treatment.

Ultimately, the success of readthrough-based therapies depends on the ability to navigate this trade-off between therapeutic restoration and proteomic disruption. Because the efficiency of both natural and drug-induced readthrough is heavily dictated by the sequence context

surrounding a stop codon, the response to treatment is not uniform across the genome. This variability creates a critical need for computational models that can decipher the "grammar" of these sequences. By accurately predicting which transcripts are most susceptible to a given drug, we can better identify viable therapeutic targets and anticipate the off-target effects that currently limit the safety of these life-saving interventions.

1.2. Objective

The primary goal of this research is to develop and validate a computational framework capable of predicting translational readthrough across diverse genomic and pharmacological contexts. By moving beyond traditional linear models, this study seeks to capture the non-linear "sequence grammar" that dictates how ribosomes interact with stop codon contexts. The specific objectives of this thesis are as follows:

- **Modelling Untreated Basal Readthrough of NTCs:** To characterise the inherent "leakiness" of natural termination codons across the human transcriptome. Predicting basal readthrough is essential for identifying genes that utilise C-terminal extensions as a standard regulatory mechanism for proteome expansion and functional diversity.
- **Predicting Drug-Dependent PTC Readthrough:** To evaluate the efficacy of nonsense suppression therapies. By modelling how specific drugs induce readthrough at pathogenic premature termination codons, I aim to help facilitate personalised treatment strategies for genetic diseases.
- **Assessing Drug-Dependent NTC Readthrough:** To quantify the potential for proteomic "off-target" effects. Modelling how therapeutic agents stimulate readthrough at healthy natural stop codons is critical for anticipating drug toxicity and the unintended production of potentially autoantigenic C-terminal extensions.
- **Mechanistic Interpretation via Ablation Studies:** To systematically investigate the learned representations of the neural network. By performing regional sequence

ablations, I aim to identify the relative contribution of upstream and downstream sequence motifs, providing biological insights into the cis-acting elements that govern readthrough.

1.3. Thesis outline

The rest of this thesis is structured into the following chapters:

- **Chapter 2: Literature Review** provides a comprehensive overview of the molecular mechanisms that modulate translational readthrough. It also summarises relevant datasets and surveys previous computational modelling approaches in the field, ranging from traditional logistic regression benchmarks to state-of-the-art transformer models.
- **Chapter 3: Methods** details the data preprocessing pipelines, and the computational modelling approaches, ranging from using pretrained model embeddings to the custom hybrid CNN-transformer architecture for readthrough prediction.
- **Chapter 4: Results** presents the performance evaluation of our model across Natural Termination Codon (NTC) and Premature Termination Codon (PTC) datasets. It features a comparative analysis against large-scale foundation models and established benchmarks, followed by systematic ablation studies and feature visualisations that map the model's learned biological priorities.
- **Chapter 5: Discussion and Conclusion** synthesises our findings, placing them within the broader context of proteome expansion and therapeutic nonsense suppression. It addresses the model's biological implications and limitations, outlines contributions to the field, and proposes future research directions for clinical and experimental validation.

In summary, this thesis project aims to build an interpretable deep learning tool to explore the phenomenon of translational readthrough. By bridging the gap between computational sequence analysis and therapeutic prediction, this work provides a new perspective on the regulation of translation and the future of precision medicine in the treatment of nonsense-mediated genetic disorders.

Chapter 2

Literature review

2.1. Mechanisms of readthrough

Translation termination normally occurs when a ribosome encounters a stop codon (UAA, UAG, and UGA) in the reading frame of the mRNA sequence. This process is regulated by a release factor complex, made up of eRF1, eRF3, and GTP, which recognises the stop codon in the ribosomal A-site and triggers the release of the nascent polypeptide chain, allowing the ribosome to be recycled for another round of translation [1, 2]. This termination process is typically very efficient and has a low error rate [3]. However, this efficiency is not absolute, and translational readthrough can occur when a near-cognate tRNA (nc-tRNA), carrying an amino acid, successfully competes with the release factors and decodes the stop codon as a sense codon. Thus, the ribosome continues translating into the 3' untranslated region (UTR), producing a C-terminally extended protein. The extent to which this "misreading" occurs is determined by a complex interplay of various factors, comprising cis-acting elements within the mRNA sequence as well as trans-acting elements present in the cellular environment.

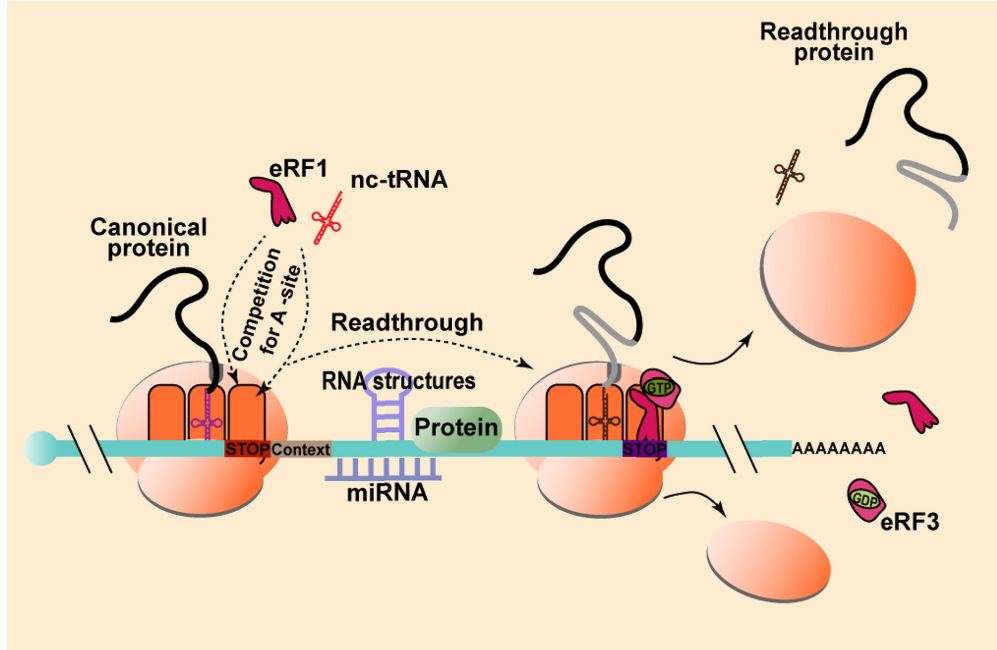


Figure 2.1: Transient molecular roadblocks: A possible mechanism for induction and regulation of SCR. During SCR, instead of eRF1, a near-cognate (nc) tRNA recognises the stop codon at the A-site of the ribosomes. This process is triggered or facilitated by several factors—local mRNA secondary or tertiary structures, RNA-binding proteins, and miRNAs. They potentially serve as transient molecular roadblocks for ribosomes at the stop codon. After SCR, the ribosomes terminate at the next available stop codon, resulting in a protein isoform with C-terminal extension.

As part of the cis-acting elements, several key sequence features, particularly those surrounding the stop codon, play a vital role in modulating readthrough efficiency. Firstly, the primary determinant of readthrough is the identity of the stop codon itself, with UGA typically being the “leakiest” or most readthrough permissive, followed by UAG and then UAA. Additionally, the nucleotides flanking the stop codon play a vital role, with the nucleotide immediately following the stop codon (+4 nt) being particularly influential. Prior studies have found that a cytosine (C) at this position helps promote readthrough by facilitating the binding of nc-tRNA, especially for UGA stop codons [6]. Beyond the +4 position, the extended sequence context, both upstream and downstream of the stop codon, further influences readthrough efficiency. For instance, across data collected from human cells, the readthrough-promoting motif UGACANNNA and the readthrough inhibitory motif UAAG(G/C)NNNC have been identified, where N stands for any given nucleotide [4]. More detailed investigations have revealed the surprising role of upstream positions (e.g., -9 and -8

relative to the stop codon) and downstream positions (+11 and +12) on readthrough levels, highlighting a complex, non-linear and context-dependent interplay among these nucleotides. Beyond the immediate stop codon neighbourhood, transcriptome-wide analyses indicate that longer-range features such as 3' UTR length, and mRNA structural motifs such as stem loops, also play a role in modulating readthrough, supporting the view that predictive models may benefit from incorporating broader mRNA context beyond the canonical stop codon context [7]. The conservation of some of these cis-acting elements, both local and distal, across species such as yeast and human cells, serves as further evidence for their fundamental role in readthrough regulation.

Thus, this inherent complexity means that the impact of individual sequence elements on readthrough is not simply additive or easily generalisable across different contexts. Instead, it is a dynamic and intricate system where various features interact to determine the overall readthrough efficiency. This makes the problem of translational readthrough particularly well suited to be tackled by machine learning approaches, such as modern neural network architectures like CNNs and transformers, which are able to capture and learn complex non-linear dependencies directly from sequence data, without the need for explicit features to be defined and manually engineered a priori.

2.1.1 Drug-induced readthrough

As discussed before, the ability of translational readthrough to restore full-length protein production from transcripts containing premature termination codons (PTCs) makes it a promising therapeutic strategy for a wide range of genetic disorders. In pursuit of this approach, several readthrough-promoting compounds have been discovered, which interfere with the translation termination process and shift the balance in favour of the nc-tRNA decoding of the stop codon.

One such class of drugs is aminoglycosides, such as G418 and gentamicin, which are well-known inducers of readthrough. They work by binding to the ribosomal A-site, causing

conformational changes that interfere with the recognition of the stop codon by release factors and enhance the binding of near-cognate tRNAs [6]. While these drugs can be effective, prior studies have found that the readthrough efficiency induced by them is highly dependent on the specific sequence context of the stop codon. For instance, in sequences treated with the drug G418, it has been found that the stop codon identity and the +4 nucleotide have an enhanced role in predicting readthrough efficiency. In addition to aminoglycosides, other compounds have also been explored for their readthrough-promoting potential. These include small molecules that directly or indirectly influence the termination machinery by depleting release factors or modulating tRNA modifications [8].

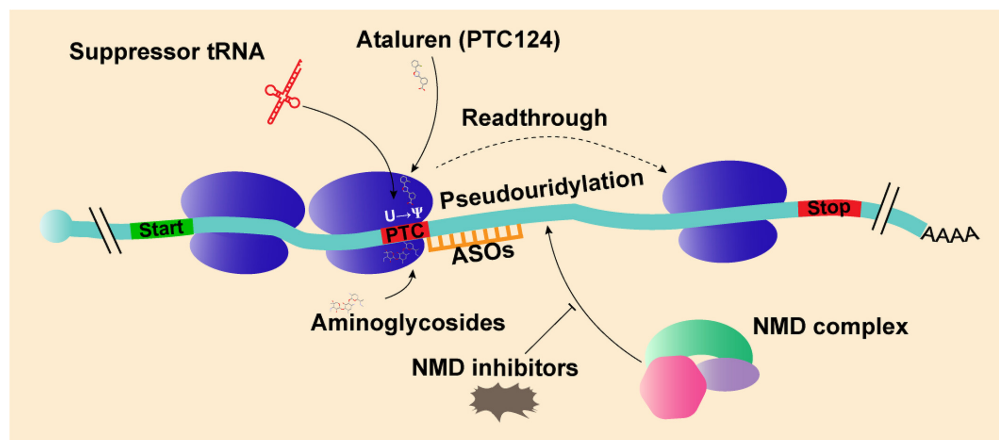


Figure 2.2: Strategies to induce SCR across disease-causing PTCs. Aminoglycosides and ataluren target ribosomes to achieve SCR. Engineered suppressor tRNAs can recognise and decode stop codons, resulting in SCR. Targeted pseudouridylation and anti-sense oligonucleotides (ASOs) can achieve transcript-specific induction of SCR. NMD inhibitors enhance the efficacy of SCR-inducing agents by increasing the cellular levels of target mRNAs with PTC.

For each of these drug classes, the readthrough efficacy can vary substantially based on the mRNA sequence context. Genome-wide experimental data from cells treated with multiple different aminoglycoside drugs heavily support the idea that the drug identity interacts with the mRNA sequence environment to predict readthrough [4]. Thus, a crucial challenge in modelling stop codon readthrough is taking into account the "sequence specificity" of drug action. Once again, neural network models are well-positioned to capture this complex interplay between sequence context and drug identity in predicting readthrough.

2.2. Available datasets

The development of robust computational models for translational readthrough requires high-quality, large-scale datasets that quantify readthrough efficiency across a vast array of sequence contexts. This study requires datasets that quantify readthrough in both Natural Termination Codons (NTCs) to understand basal regulation and Premature Termination Codons (PTCs) to evaluate therapeutic potential. Based on a review of the literature, two primary datasets were selected.

2.2.1 Toledano Dataset

The Toledano study aimed to systematically quantify how small molecules induce readthrough across a comprehensive library of pathogenic mutations.

- **Experimental Assay:** The authors utilised a fluorescence reporter-based assay integrated into the genome of HEK293T cells. The construct consisted of a dual-fluorescence reporter where a stop codon (either an NTC or a disease-causing PTC) was placed between a Green Fluorescent Protein (GFP) and a mCherry fluorescent protein.
- **Calculating Efficiency:** Readthrough efficiency was calculated based on the ratio of mCherry to GFP fluorescence. Because GFP is expressed before the stop codon and mCherry only after a successful readthrough event, this ratio directly quantifies the frequency at which ribosomes bypass the termination signal.
- **Toledano PTC Data:** This library includes approximately 5,800 human disease-causing PTCs sourced from the ClinVar database and cancer variant datasets.
- **Toledano NTC Data:** In addition to the PTC library, the researchers evaluated endogenous NTC sequences genome-wide. They measured readthrough for five of the best-performing drugs from the initial screen, providing a crucial dataset for evaluating the "off-target" effects of nonsense suppression therapies.

- **Key Findings:** A central finding was that drug-induced readthrough is highly idiosyncratic; different compounds preferentially promote readthrough of distinct and complementary subsets of PTCs depending on the local nucleotide environment.

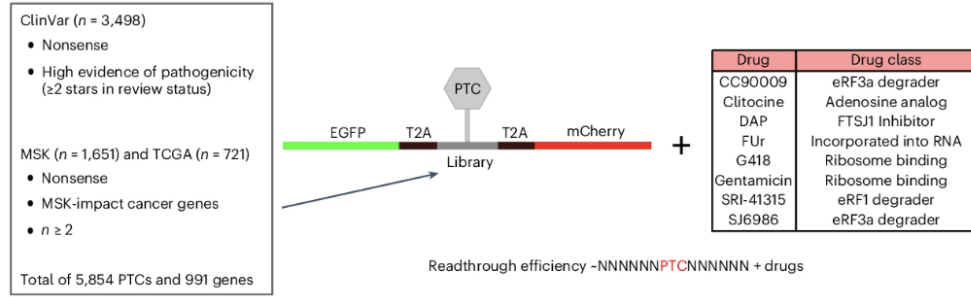


Figure 2.3: Readthrough drugs stimulate full-length protein synthesis and decrease NMD-mediated transcript degradation. 5,800 nonsense variants in human genetic diseases and cancer were retrieved from ClinVar, TCGA and MSK-IMPACT datasets, cloned in a readthrough reporter, integrated into the genome of HEK293T_LP human cell line and treated with eight readthrough compounds. A readthrough efficiency value was obtained for each variant–drug pair.

2.2.2 Wangen Dataset

The Wangen dataset [10] provides a different perspective by utilising transcriptome-wide measurements of readthrough at endogenous NTCs in their native genomic contexts.

- **Experimental Assay:** This study employed ribosome profiling (Ribo-seq) in HEK293T cells. Ribo-seq involves deep sequencing of mRNA fragments protected by ribosomes, providing a high-resolution map of ribosome positions across the entire transcriptome.
- **Calculating RRTS:** Readthrough efficiency was quantified using the Ribosome Readthrough Score (RRTS). The RRTS is calculated as the ratio of ribosome density in the region immediately following the stop codon (the proximal 3' UTR) to the ribosome density within the coding sequence (CDS). A higher RRTS indicates a greater proportion of ribosomes continuing translation past the NTC.

- **Wangen NTC Data:** The authors quantified readthrough efficiency at NTCs for untreated cells and cells exposed to various aminoglycosides (such as G418 and Gentamicin) under multiple treatment durations and concentrations.
- **Key Findings:** Their results demonstrated that while the immediate stop codon context is a primary determinant, broader transcript-level features—such as the length of the 3' UTR and specific mRNA structural motifs—significantly modulate readthrough efficiency under drug-stimulated conditions.

Drug Shorthand	Treatment Details
G418_500	G418 at 500 $\mu\text{g}/\text{mL}$
G418_2000	G418 at 2000 $\mu\text{g}/\text{mL}$
G418_500_10min	G418 at 500 $\mu\text{g}/\text{mL}$ for 10 minutes
amik	Amikacin
genta	Gentamicin
neo	Neomycin
parom	Paromomycin
tobra	Tobramycin
untr	Untreated (basal readthrough)

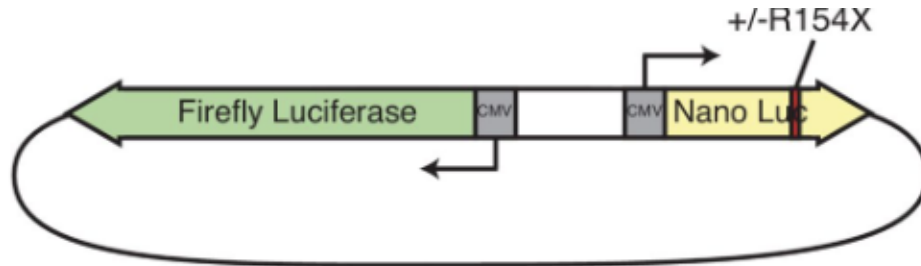


Figure 2.4: Schematic of reporter constructs used in this study. Full-length firefly luciferase (FLuc) is expressed from the reverse strand, and nano luciferase (NLuc) encoding either an arginine or a UGA stop codon at amino acid 154 is expressed from the forward strand.

2.3. Previous computational models

Before proceeding with the development of a novel neural network architecture, it is essential to review existing methodologies that have attempted to decode the sequence-to-function

relationship of translational readthrough. Prior studies primarily utilise traditional machine learning frameworks that require the explicit definition of sequence-related features relevant to readthrough manually. The following contains an overview of three key approaches.

2.3.1 Logistic Regression Model

Toledano et al. established the current state-of-the-art benchmark by developing interpretable predictive models to estimate drug-induced readthrough.

- **Methodology:** The researchers employed a regularised linear regression framework. The models were built using manually engineered features, including stop codon identity (UAA, UAG, UGA), the identity of the nucleotide at the +4 position, and additional upstream and downstream sequence descriptors.
- **Application:** Separate models were trained on the **Toledano PTC data** and **Toledano NTC data**.
- **Results and Benchmarks:** The model demonstrated that local sequence context contains a substantial fraction of the predictive signal, achieving a pan-drug correlation of $R^2 = 0.83$. For specific drug treatments, the R^2 values ranged from 0.89 to 0.55 for PTCs. These values serve as the primary baseline for assessing the performance of the hybrid model on the same dataset.

2.3.2 Random Forest Model

Schueren et al. explored the use of ensemble learning to identify broader regulatory motifs beyond simple linear combinations of nucleotides.

- **Methodology:** A Random Forest (RF) classifier was implemented to capture non-linear interactions between various cis-acting elements. This approach was instrumental in identifying the impact of sequence context on readthrough efficiency.

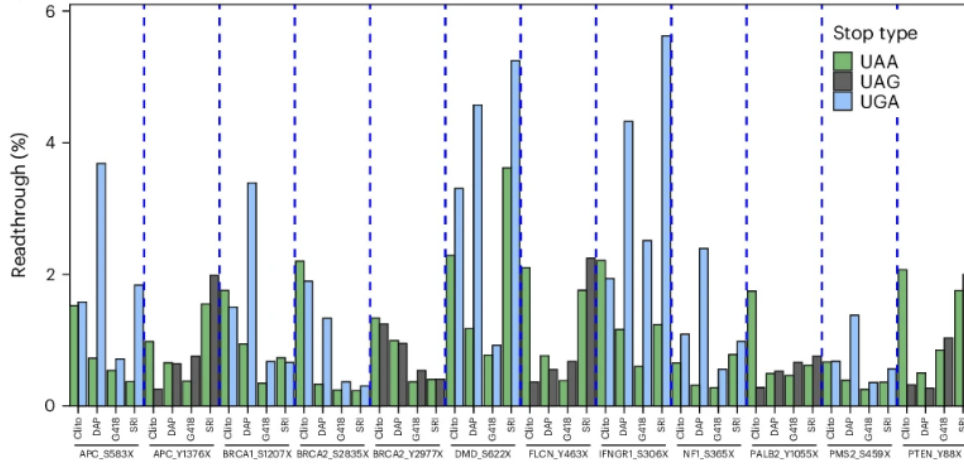


Figure 2.5: Readthrough efficiency for 12 multistop variants across four drugs. Each multistop variant comprises two different nonsense mutations (different stop codon identities) observed in the same genomic locus.

- **Application:** This modelling was primarily applied to endogenous sequences, focusing on identifying which biological features distinguish "leaky" stop codons from "tight" ones.
- **Results:** The RF model successfully highlighted that distal features and mRNA structural motifs significantly modulate readthrough. The results obtained, specifically the per-drug AUROC values, act as a benchmark for the computational models.

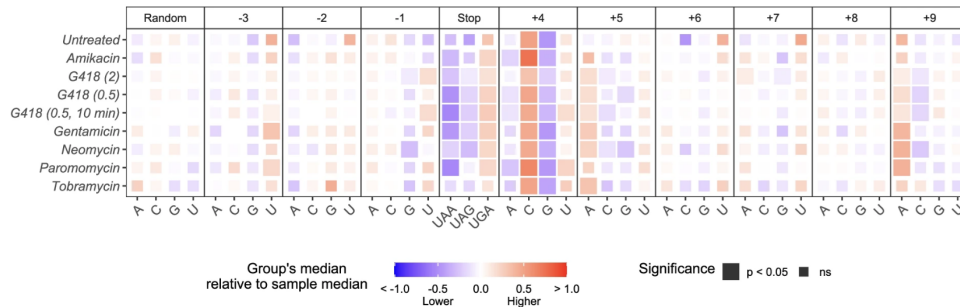


Figure 2.6: Differences between median readthrough efficiency of all mRNAs in the sample and median readthrough efficiency of a group of mRNAs containing a particular stop codon or nucleotide.

2.3.3 TransStop

TransStop [17], a preprint released concurrently, represents a key innovation in the field, utilising a transformer-based genomic language model (gLM) to predict translational readthrough efficacy.

- **Methodology:** TransStop was fine-tuned from the pre-trained “**nucleotide-transformer-v2-500m-multi-species**” foundation model (described in more detail in Sec. 2.4.2). It incorporates a unique **cross-attention mechanism** where a learned drug embedding (query) attends to the sequence embeddings (keys and values). This allows the model to dynamically weigh different parts of the sequence context based on the specific drug being modelled.
- **Application:** The model was trained and optimised primarily on the **Toledano PTC data**.
- **Results:** TransStop achieved a remarkable global $R^2 = 0.94$ on a held-out test set, systematically outperforming the Toledano logistic regression model across every tested condition. The improvement was most pronounced for eRF1/eRF3 inhibitors like SJ6986 (0.91 vs. 0.71) and CC90009 (0.79 vs. 0.55).
- **Biological Interpretation:** Visualisations of the model’s latent space using UMAP revealed that TransStop spontaneously organises data based on biological principles, such as clustering sequences by stop codon identity (UAA, UAG, and UGA), despite having no explicit supervision to do so.

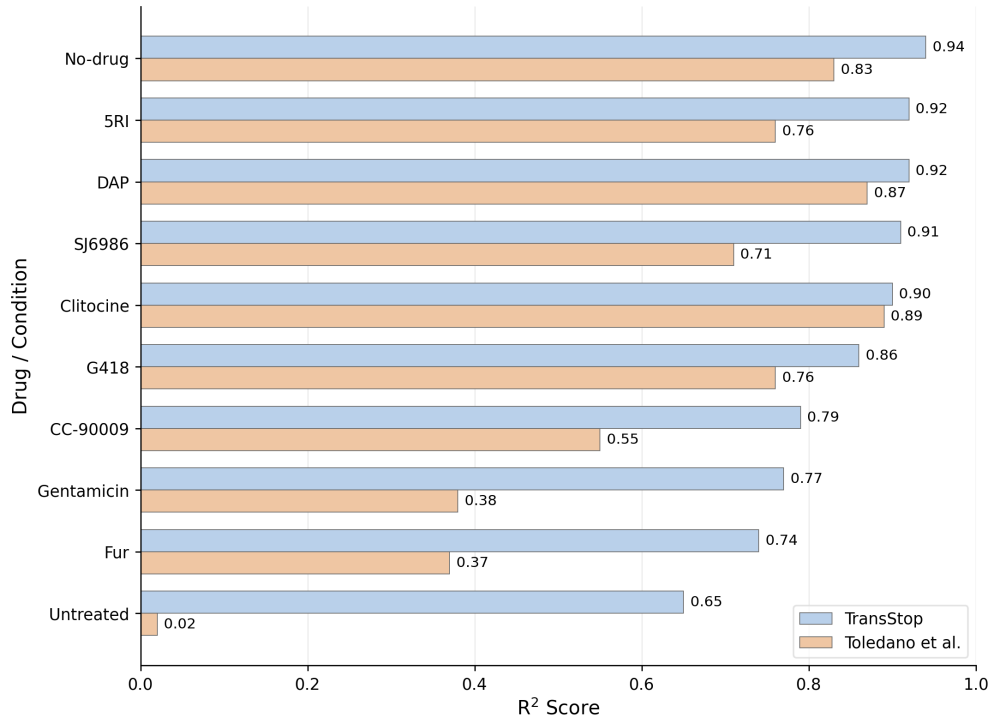


Figure 2.7: Bar chart of performance (R^2 scores) for the TransStop model (blue) and the logistic regression model from Toledano et al. (orange) for each drug and for the overall “Pan-drug” performance.

2.4. Relevant transformer models

In this section, I describe the deep learning architectures used to model nucleotide sequences. Prior work has adapted transformer-based architectures, originally developed for natural language, to a range of downstream classification and regression tasks on genomic sequences.

2.4.1 Language modelling with transformers

The transformer [11] is a deep neural network for processing sequences and has become the state of the art across most natural language tasks. A sentence is first broken into discrete units called tokens, each of which is mapped to a high-dimensional vector known as an embedding. The transformer then processes these embeddings in parallel, allowing each position to selectively incorporate information from every other position through a mechanism

called attention. After several such layers, each embedding captures global sequence context and can serve as input for various downstream classification or regression tasks.

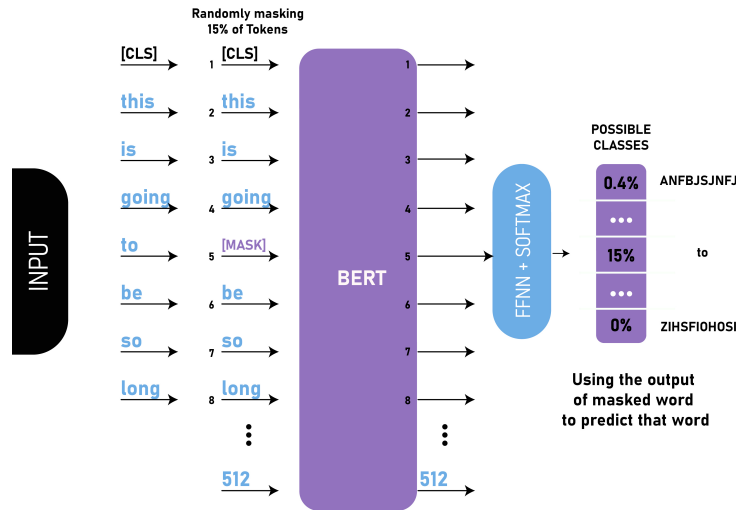


Figure 2.8: Schematic diagram of the BERT architecture. ([image source](#))

A widely used transformer model is BERT [12], which is pretrained by randomly masking tokens in the input sequence and training the model to predict the masked identities from their surrounding context (see Fig. 2.8). This objective forces the learned embeddings to encode semantically meaningful relationships between tokens. For downstream tasks, a sequence-level representation is typically obtained either by averaging the final-layer embeddings or by using the embedding of a special [CLS] token prepended to the sequence.

In the genomic setting, tokens correspond to nucleotides or short k-mers, and sequence context plays an analogous role to surrounding word context in natural language.

2.4.2 Nucleotide Transformer

The nucleotide transformer [13] adapts BERT for the genomic setting. The nucleotide sequences are broken down into 6-mers (the tokens), with the vocabulary consisting of all $4^6 = 4,096$ possible 6-mer combinations over $\{A, T, C, G\}$, together with a few special tokens.

Architecturally, the model is similar to BERT (Fig. 2.9): an embedding layer and a stack

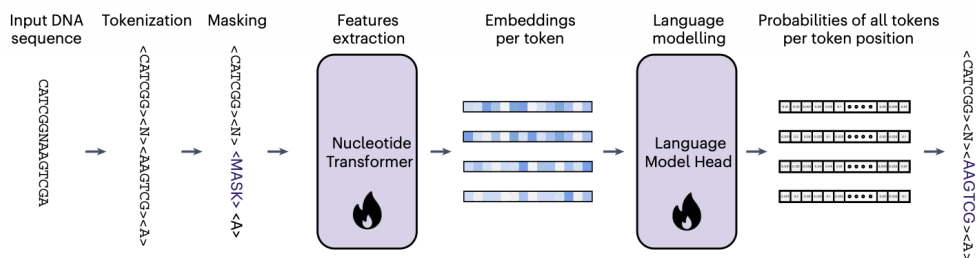


Figure 2.9: Pretraining the nucleotide transformer. ([image source](#))

of self-attention blocks in which every token attends to every other token in the window. Four first-generation models were released, ranging from 500 million to 2.5 billion parameters, differing in the depth and width of the transformer stack but sharing the same tokenisation and objective.

Pretraining uses the masked language modelling objective of BERT: roughly 15% of tokens in each input window are masked, and the model is trained to predict the original 6-mer at each masked position from the surrounding sequence context. Three pretraining corpora of increasing biological diversity were used: (i) the human reference genome GRCh38 alone ($\sim 3.2 \times 10^9$ nucleotides); (ii) high-coverage genomes of 3202 individuals from the 1000 Genomes Project [14]; and (iii) a multispecies corpus of 850 reference genomes sampled across the NCBI RefSeq collection.

The paper found that the multispecies corpus turns out to yield stronger representations for *human* downstream tasks than either human-reference-only or 1000 Genomes pretraining, despite containing no additional human sequence. A plausible interpretation is that the masked-token prediction implicitly rewards the model for recognising sequence patterns that are conserved across evolutionary distance. Probing analyses in the same paper further suggest that the learned representations carry substantial biological structure: this motivates using the architecture for readthrough prediction.

2.4.3 mRNA-LM

mRNA-LM [15] takes a different approach: rather than training a single language model over the full nucleotide sequence, it trains three smaller BERT-style models, one for each of the functionally distinct regions of an mRNA: the 5' untranslated region (UTR), the coding sequence (CDS), and the 3' UTR. It then integrates their representations into a joint full-length model. This design is motivated by the observation that the three regions play biologically distinct roles.

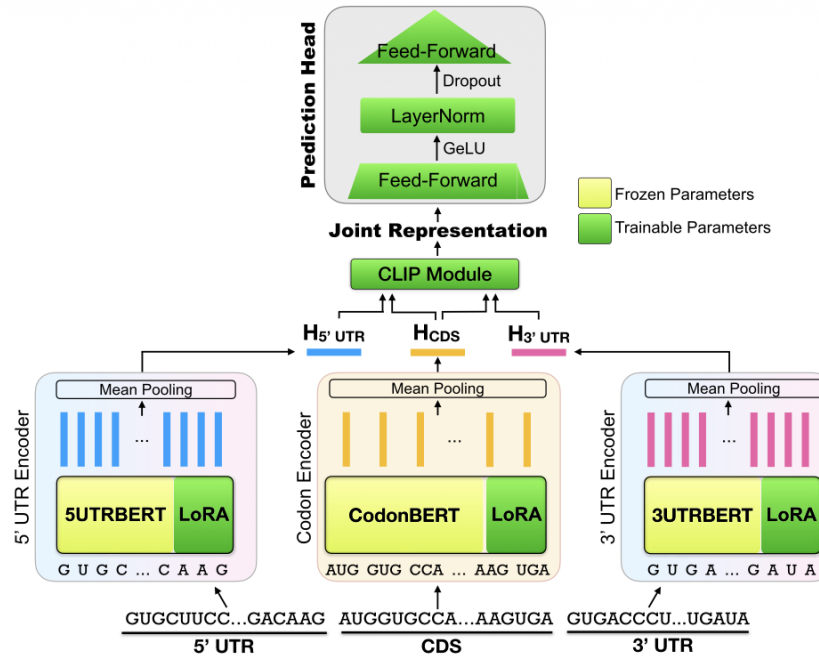


Figure 2.10: Overview of mRNA-LM. Three BERT models are pretrained separately on the three mRNA regions, then aligned via a CLIP module and fine-tuned jointly for downstream prediction tasks. ([image source](#))

Each of the three submodels is a separate encoder. The CDS encoder, CodonBERT [16], is codon-based: each of the 64 possible three-nucleotide codons is a distinct token, reflecting the fact that codons are the functional units of translation. The two UTR encoders, 5'UTRBERT and 3'UTRBERT, are nucleotide-based, treating each nucleotide directly as a token. All three models are pretrained on roughly 10 million full-length mRNA sequences drawn from mammals, yeast, bacteria, and human viruses, using the standard masked language modelling

objective with 15% of tokens masked per input.

To produce a full-length mRNA representation, the token embeddings from each of the three pretrained submodels are averaged into a single embedding, and these three segment embeddings are aligned in a shared latent space using a contrastive language-image pretraining (CLIP) module [18]. Originally developed to align image and text embeddings, CLIP here treats the three mRNA regions as distinct modalities to align them. The aligned embeddings are concatenated and passed through a small prediction head for downstream tasks.

For downstream tasks, the pretrained submodels are kept frozen and updated only through low-rank adaptation (LoRA) [19], a technique which inserts small trainable rank-decomposition matrices into each transformer layer.

An interesting result is that the full three-region mRNA-LM model outperforms each of its constituent submodels on every downstream task tested, confirming that UTR and CDS representations carry complementary information. However, replacing the CLIP module with a simpler concatenation of the three segment embeddings produced only modestly worse results, suggesting that most of the benefit comes from modelling all three regions jointly rather than from the CLIP alignment itself.

Chapter 3

Methods

This chapter details the computational frameworks and experimental methodologies employed to predict translational readthrough across the Toledano and Wangen datasets. The chapter begins with the dataset preprocessing pipelines, including the extraction of canonical transcript sequences and the transformation of experimental scores for machine learning tasks. Section 3.2 defines the specific regression and classification tasks addressed, along with the performance metrics used for evaluation. I then describe the baseline models in Section 3.3 and Section 3.4, which leverage frozen embeddings from the large-scale pre-trained Nucleotide Transformer and mRNA-LM architectures. Finally, Section 3.5 introduces a novel, custom hybrid CNN-transformer architecture designed to capture local regulatory patterns, followed by the specific evaluation protocols for the Toledano NTC, PTC, and Wangen transcriptome-wide datasets.

3.1. Dataset Preprocessing

The preparation of genomic data for deep learning requires the conversion of raw experimental results and biological sequences into a standardised format. The following sections detail the preprocessing pipelines for the two primary datasets used in this study.

3.1.1 Toledano Dataset Preprocessing

The preprocessing of the Toledano dataset was relatively straightforward, as the authors provided the data in a pre-compiled format. The primary data was obtained as a downloadable CSV file containing approximately 5,800 instances for the PTC dataset and approximately 18,000 instances for the NTC dataset. Each instance in the file consisted of:

- **Sequence Context:** A 132-nucleotide window centred around the stop codon.
- **Readthrough Efficiency:** A continuous numerical value representing the measured efficiency for each sequence-drug combination.

The sequences were used directly for training the local hybrid model, while the target readthrough efficiency values were $\log(x+1)$ -transformed to stabilise variance during regression training.

3.1.2 Wangen Dataset Preprocessing

The preprocessing of the Wangen NTC data was significantly more complex, as the original study provided transcript-level identifiers rather than fixed-length sequence windows. To prepare this data for modelling, the following pipeline was implemented:

- **Sequence Extraction:** Using the Gene IDs provided by the authors, I extracted the full Coding Sequence (CDS) and 3' Untranslated Region (UTR) for each entry from the

Ensembl database. In cases where multiple isoforms existed, the canonical transcript variant was consistently selected to ensure biological relevance and data consistency.

- **Target Transformation:** The original dataset quantified readthrough using a Ribosome Readthrough Score (RRTS). However, initial analysis revealed that the RRTS was zero for the majority of transcripts, creating a highly sparse distribution unsuitable for regression.
- **Binary Classification Mapping:** To mitigate the sparsity of the RRTS, the task was reformulated as a binary classification problem. Transcripts with an RRTS greater than zero were labelled as "induced readthrough" (positive class), while those with an RRTS of zero or less were labelled as "not induced" (negative class).

3.2. Metrics and Evaluation

I tackle the following tasks:

- **Regression** of continuous readthrough efficiency on both the NTC and PTC Toledano datasets.
- **Classification** of binary readthrough on the NTC Wangen dataset.

For the above tasks, I use the following standard metrics:

R²: For regression, R^2 gives us a measure of the degree of variance in the target variable explained by the model. It is given by:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

where SS_{res} is the sum of the squared differences between the observed values and model predictions, and SS_{tot} is the sum of the squared differences between the observed values and the mean of all observed values.

Spearman ρ : For regression, Spearman’s rank correlation coefficient ρ quantifies whether higher (lower) values in the experimental dataset correspond to higher (lower) model predictions.

AUROC. For classification, the AUROC (Area Under the Receiver Operating Characteristic) is the area under the TPR-FPR curve, and measures the probability that the model assigns a higher score to a positive instance, if a positive and negative instance are randomly selected.

3.3. Nucleotide Transformer

For both Nucleotide Transformer and mRNA-LM, I evaluated whether large-scale pretraining yields representations rich enough to transfer to readthrough prediction. I froze the pretrained weights and trained a classification or regression head on top of the extracted embeddings.

For the Nucleotide Transformer, each sequence is tokenised into 6-mers and passed through the frozen model. I used the [CLS] token embedding (1024 dimensions) from the final hidden layer as the sequence representation, and used XGBoost [21], a tree-based boosting algorithm, as a regressor. To set the hyperparameters and obtain evaluation metrics, I performed 5-fold cross-validation (the dataset is divided into 5 splits, and over the course of 5 iterations, 4 are chosen as the training splits and 1 is chosen as the test split; the final metric is the average over all iterations). To select the values of hyperparameters, I used Optuna [22], an automatic hyperparameter optimisation framework. For classification, most of the design choices stayed the same, except I used an XGBoost classifier with weights to handle class imbalance.

3.4. mRNA-LM

Just like the Nucleotide Transformer, I froze the weights of the mRNA-LM. I obtained embeddings by passing the CDS sequence through CodonBERT and the 3’UTR sequence through UTR-BERT. Then, to obtain a single embedding per region, I mean-pooled the

final hidden states over all content tokens, excluding the [CLS] and [SEP] positions. The two 768-dimensional vectors are concatenated to form a 1536-dimensional embedding per sequence. I opted for mean pooling instead of using the CLIP module, for simplicity, and because the original paper did not show significant improvements when using CLIP. Note that the mRNA-LM is pretrained on full-length CDS and UTR sequences, whereas the Toledano dataset comprises only 132 nt of sequence context around the stop codon; I therefore do not use this on the Toledano dataset.

3.5. A hybrid CNN–transformer for local readthrough prediction

Given the limited sizes of the Toledano and Wangen datasets, a large transformer trained from scratch would overfit heavily. Empirically, I also found that fine-tuning or reusing frozen embeddings from foundation models underperformed. Therefore, I designed a compact model, trained from scratch, whose architecture encodes as much prior knowledge about the readthrough problem as possible (see Fig. 3.1). This approach is motivated by two observations about the task.

First, readthrough is a fundamentally *local* phenomenon. The nucleotides immediately flanking the stop codon are the dominant determinants of readthrough efficiency, and distant sequence context offers little additional signal. Second, a standard transformer with full self-attention allows every nucleotide in the input window to incorporate information from every other nucleotide, regardless of distance; with only a few thousand training examples, this flexibility gives the model far more opportunities to fit spurious long-range correlations than to learn genuine local regulatory patterns.

These motivate the following architectural choices, each encoding a specific inductive bias:

- **Convolutional front end.** Before any attention is applied, the input passes through one-dimensional convolutional layers. A convolution slides a small filter over adjacent

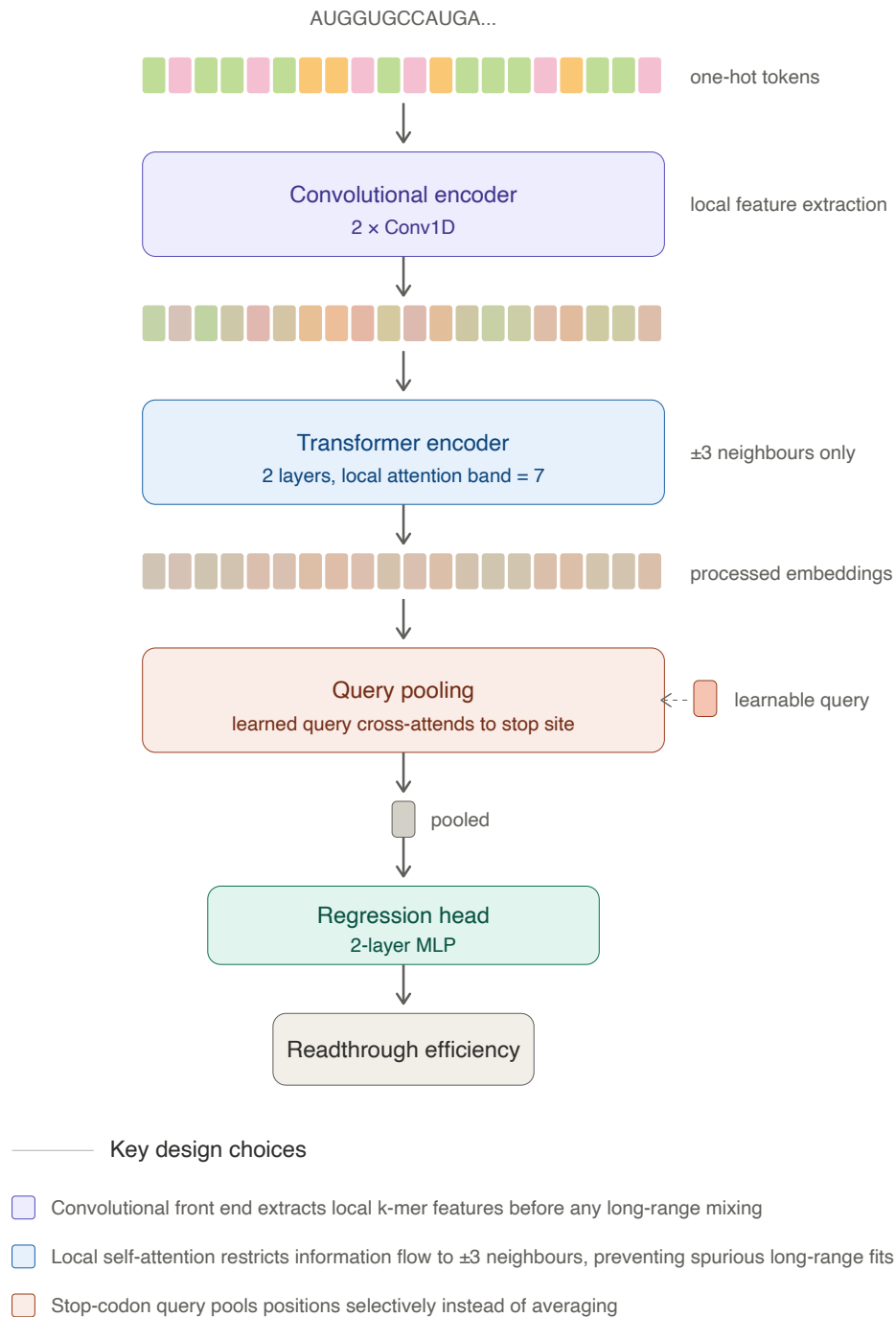


Figure 3.1: Schematic diagram for the hybrid CNN-transformer architecture for readthrough prediction. Nucleotide tokens pass through two convolutional layers, two transformer layers with local self-attention, and a learned stop-codon query that cross-attends over the sequence to produce a single pooled representation. The final pooled token is either fed to a classification or regression head.

nucleotides, producing at each position a feature vector that summarises its local neighbourhood. This is the same inductive bias that has made convolutional architectures a common choice for small-to-medium-scale tasks, and more broadly, convolutional models outperform transformers trained from scratch when training data is limited [20]. The convolutional layers commit the model to summarising local sequence context before any long-range interactions are modelled.

- **Local self-attention.** The transformer layers that follow the convolutions retain attention’s ability to let neighbouring positions exchange information, but restrict this exchange to a small band: each nucleotide attends only to its three nearest neighbours on either side, rather than to the entire window. This structurally prevents the model from fitting long-range spurious correlations while preserving attention’s flexibility for modelling local interactions that a pure convolutional stack would find harder to express; for instance, interactions between the stop codon and the +4 position that depend on the identity of both.
- **Stop-codon-centered readout.** Rather than averaging the per-position outputs of the transformer into a single sequence representation, which would weight every position equally, the model uses a learned ”query” vector that attends over the transformer outputs and selectively pools the positions most relevant to readthrough.

3.5.1 Evaluation on Toledano data

As described earlier, the Toledano dataset contains $\sim 18,000$ sequences measured across 5 readthrough-inducing drugs, each with a continuous efficiency score. Because each drug induces readthrough through a different mechanism and to different degrees, I trained a separate regression model per drug for simplicity. Each model uses the shared backbone described in Sec. 3.5, with a two-layer MLP regression head that outputs a single scalar prediction.

Each sequence in the dataset is 132 nucleotides long, with the stop codon at a fixed

position (index 63). From this, a symmetric window of $\pm k$ nucleotides around the stop codon is extracted, yielding an input of $2k+3$ tokens (including the three stop-codon nucleotides). To determine how much flanking context the model needs, I swept k over $\{10, 15, 20, 30, 45, 63\}$ nucleotides. In addition to the nucleotide tokens, each position receives a *stop-codon-relative positional encoding*: an integer index equal to its signed distance from the first nucleotide of the stop codon. This is embedded via a learned lookup table and added to the nucleotide embedding, giving the model an explicit representation of where each nucleotide sits relative to the stop codon.

As controls, I ran two ablation experiments at the ± 10 nt window to assess the relative contributions of upstream (CDS) and downstream (3' UTR) context. The CDS-only ablation retains only the 10 nucleotides upstream of the stop codon plus the stop codon itself; the UTR-only ablation retains only the stop codon plus 10 downstream nucleotides. I also ran a shuffled-sequence control in which the nucleotide tokens within each window are randomly permuted while the positional encodings are held fixed, destroying sequence information while preserving the overall nucleotide composition and positional structure.

3.5.2 Evaluation on the Toledano PTC dataset

Since the PTC dataset (with 5800 sequences in total) is smaller than the NTC dataset (with 5800 sequences per drug), I opt to use a single pan-drug regressor for the PTC dataset. For details on the architecture, see Sec. 3.5.3.

3.5.3 Evaluation on Wangen data

The Wangen dataset presents a different modelling challenge. Rather than measuring continuous readthrough efficiency under a single compound, it reports binary readthrough calls across nine drugs. To account for data scarcity and to prevent wasting shared signals, I train a single *pan-drug* classifier that shares sequence-level feature extraction across drugs while conditioning its predictions on drug identity.

The model uses the same convolutional and local-attention backbone described above, with three modifications to support drug-conditioned binary classification:

- **Drug embedding.** Each of the nine drugs is assigned a learned 32-dimensional embedding vector. This embedding is projected to match the nucleotide embedding dimension element-wise to every position in the input, so the convolutional and transformer layers can learn drug-specific sequence patterns.
- **Drug-conditioned query pooling.** The learned query vector used for pooling is conditioned on the drug embedding before cross-attending over the transformer outputs, giving each drug its own readout perspective over the sequence representation.
- **Binary classification head.** The regression head is replaced with a two-layer MLP that outputs a single logit, trained with binary cross-entropy loss.

The prevalence of the class varies substantially between drugs, from 66% positive for G418 at 2000 $\mu\text{g/ml}$ to 9.7% for the untreated condition. To prevent the model from collapsing to the majority class for low-prevalence drugs, I applied two forms of balancing. First, each training sample receives a weight inversely proportional to its drug-specific class prevalence, so that rare positives (e.g. untreated readthrough events) contribute more to the loss. Second, minibatches are assembled using a weighted random sampler that ensures balanced representation across drug-label combinations within each batch.

Chapter 4

Results

In this chapter, I present the performance of various computational architectures in predicting translational readthrough across the Toledano and Wangen datasets. I begin by evaluating frozen embeddings from two state-of-the-art genomic foundation models—the Nucleotide Transformer and mRNA-LM. Following these baselines, I present the results of my custom hybrid CNN-transformer model, exploring the impact of varying sequence context windows on predictive accuracy. Finally, I detail the model’s performance on both continuous readthrough efficiency (Toledano) and binary readthrough classification (Wangen), with comparative analysis against existing linear benchmarks.

4.1. Pretrained models

To establish a performance baseline, I first evaluated whether large-scale genomic language models (gLMs) could capture the intricate sequence dependencies governing translational readthrough. These models are typically pretrained on millions of genomic sequences to learn general structural and evolutionary patterns. My objective was to determine if these high-dimensional representations—developed for general genomic tasks—could be transferred to the specific problem of readthrough prediction with minimal fine-tuning.

4.1.1 Nucleotide transformer model

This section outlines the performance of the Nucleotide Transformer (v2-500m) when used as a frozen feature extractor. I evaluated whether the embeddings generated from this large-scale foundation model could be transferred to predict readthrough efficiency for both the Toledano and Wangen datasets.

Toledano NTC data

The following table summarises the performance on the Toledano NTC dataset. As described earlier, I utilise R^2 to evaluate the regression model’s ability to exactly predict the readthrough efficiency across drug conditions. I also use Spearman’s correlation coefficient ρ to quantify how directionally correlated the model’s predictions and the experimental values are.

Treatment	R^2	Spearman’s ρ
Clitocine	0.0296	0.16
DAP	0.0867	0.30
G418	0.0393	0.19
SJ6986	0.0819	0.28
SRI	0.0516	0.26

Table 4.1: Model performance across different treatments.

The results in Table 4.1 indicate that the R^2 values remain near zero across all drug treatments, demonstrating that the frozen foundation model embeddings cannot accurately predict the absolute value of readthrough efficiency. However, the Spearman correlation coefficients (ρ) are consistently greater than 0. This suggests that while the model struggles with precise regression predictions, the sequences contain *some* biological signals that allow the model to rank the relative efficiency of different sequence contexts.

Toledano PTC data

Performance on the Toledano PTC dataset followed a similar trend. I evaluated the model across nine distinct conditions using the coefficient of determination (R^2) and Spearman’s rank correlation (ρ).

Treatment	R^2	Spearman’s ρ
FUr	0.0075	0.00
Gentamicin	0.0011	0.01
CC90009	0.0106	0.08
G418	0.0148	0.12
Clitocine	0.0378	0.17
DAP	0.0209	0.16
SJ6986	0.0384	0.20
SRI	0.0270	0.19
Untreated	0.0310	0.12

Table 4.2: Model performance across treatments.

As shown in Table 4.2, the R^2 values remain negligible across all pharmacological treatments. This confirms that the frozen foundation model embeddings are unable to predict the exact magnitude of readthrough efficiency induced by these compounds. As before, the Spearman correlation exceeds 0, reinforcing the conclusion that while the model fails at precise regression, it captures some underlying sequence signals for readthrough prediction.

Wangen NTC data

The model was evaluated on the transcriptome-wide Wangen NTC dataset. Because the target values in this dataset were reformulated into a binary classification task as described in the methodology, performance was quantified using the AUROC to assess the model’s ability to distinguish between induced and non-induced readthrough events.

As shown in Table 4.3, the AUROC values for the Wangen NTC data range from 0.565 to 0.622. These results remain consistent with the findings from the Toledano datasets and indicate that while frozen embeddings from the Nucleotide Transformer perform slightly above

Drug	AUROC
G418_2000	0.565
G418_500	0.596
G418_500_10min	0.576
amik	0.614
genta	0.621
neo	0.608
parom	0.579
tobra	0.616
untr	0.622

Table 4.3: AUROC across drugs.

random chance, they fall short of the high-confidence classification required for biological discovery. This further highlights that general-purpose genomic representations, while useful for capturing broad sequence-related features, lack the specific local resolution needed to accurately predict translational readthrough across diverse treatment conditions.

4.1.2 mRNA-LM model

As described in the methodology, extracting embeddings from the mRNA-LM model requires full CDS and 3'UTR sequences to provide a complete transcriptomic context. Hence, the model could not be evaluated on the Toledano datasets, which consisted only of a restricted 132-nucleotide sequence window surrounding the stop codon.

Wangen NTC data

The mRNA-LM model was evaluated on the Wangen NTC data to determine if a multi-region language model—explicitly designed to process functionally distinct mRNA segments—could be leveraged to predict the occurrence of translational readthrough.

Drug	AUROC
G418_2000	0.478
G418_500	0.528
G418_500_10min	0.523
amik	0.570
genta	0.568
neo	0.546
parom	0.524
tobra	0.551
untr	0.570

Table 4.4: Readthrough classification across drugs (mRNA-LM).

As shown in Table 4.4, the classification performance using frozen mRNA-LM embeddings is notably poor, with AUROC values hovering near the 0.50 baseline for random chance. In some instances, such as G418_2000, the performance even falls slightly below this baseline. These results suggest that despite the model’s architectural focus on mRNA structure, the frozen representations do not capture the specific local interactions necessary to resolve translational readthrough efficiency.

This reinforces the earlier observation that large-scale foundation models, even when architecturally tailored to mRNA regions, remain biased toward global sequence properties.

Without targeted fine-tuning on large-scale readthrough-specific datasets, these models are unable to distinguish the sparse regulatory signals that differentiate readthrough-permissive transcripts from non-permissive ones.

4.1.3 Summary

Overall, the evaluation of frozen embeddings from the Nucleotide Transformer and mRNA-LM models demonstrated poor performance in predicting translational readthrough. This lack of predictive power is likely due to a significant mismatch between the scale of these foundation models and the relatively small experimental datasets used for downstream training.

Both the Nucleotide Transformer and mRNA-LM are pretrained on millions of diverse genomic sequences, which allows their embeddings to capture broad evolutionary and structural properties of DNA and mRNA. However, the specific regulatory "grammar" of translational readthrough represents a sparse signal within this high-dimensional space. Because the available experimental data is limited in scale, the downstream classifiers remain biased toward the general sequence properties learned during pretraining. Consequently, these models struggle to extract the specific, high-resolution features relevant to readthrough, highlighting the need for biologically-constrained, custom architectures like the one presented in the following sections.

4.2. Hybrid CNN transformer model

The following section details the results obtained using the custom hybrid CNN-transformer architecture. This model was designed to overcome the limitations of large-scale pretrained models by utilising a more compact scale model specifically optimised for the available experimental datasets.

4.2.1 Toledano NTC data

To determine the spatial extent of the regulatory signals governing readthrough, I tested multiple sequence context windows around the stop codon. For each drug treatment, the predictive performance was quantified using the coefficient of determination (R^2). These results were then compared against the R^2 values obtained by the benchmark logistic regression model developed by Toledano et al.[\[9\]](#). The performance metrics across varying window sizes are summarised in Table [4.5](#).

Drug	$\pm 10\text{nt}$	$\pm 15\text{nt}$	$\pm 20\text{nt}$	$\pm 30\text{nt}$	$\pm 45\text{nt}$	Full sequence	Benchmark Paper
Clitocine	0.717	0.717	0.717	0.717	0.713	0.707	0.720
DAP	0.889	0.889	0.890	0.892	0.894	0.893	0.880
G418	0.605	0.616	0.624	0.627	0.630	0.636	0.660
SJ6986	0.536	0.540	0.548	0.563	0.583	0.603	0.520
SRI	0.752	0.753	0.762	0.760	0.758	0.737	0.700

Table 4.5: Performance across different context window sizes (transposed).

Several key findings emerge from this analysis:

- **Context Sensitivity:** For the majority of tested drugs, predictive accuracy increases incrementally as the sequence context around the stop codon is expanded.
- **Clitocine Specialisation:** Performance for Clitocine remains largely stagnant and shows a slight decline with the largest windows. This indicates that Clitocine-induced readthrough is primarily governed by the immediate local sequence context, and wider windows likely introduce detrimental noise into the model.
- **Performance vs. Benchmark:** The hybrid model matches or exceeds the prediction accuracy of the benchmark logistic regression paper in several instances. This is achieved without the need for manual feature engineering, as the model implicitly learns relevant sequence features directly from the data.
- **Selection of Optimal Model:** While drug-specific requirements vary, a context window of approximately ± 45 nt consistently provides the best results across all drugs on average. Consequently, this window size was selected as the optimal model configuration for all subsequent ablation studies and analyses.

To further illustrate the model’s accuracy, the scatter plot in Fig. 4.1 provides a visualisation of the correlation between the experimentally measured readthrough efficiency and the values predicted by the hybrid model.

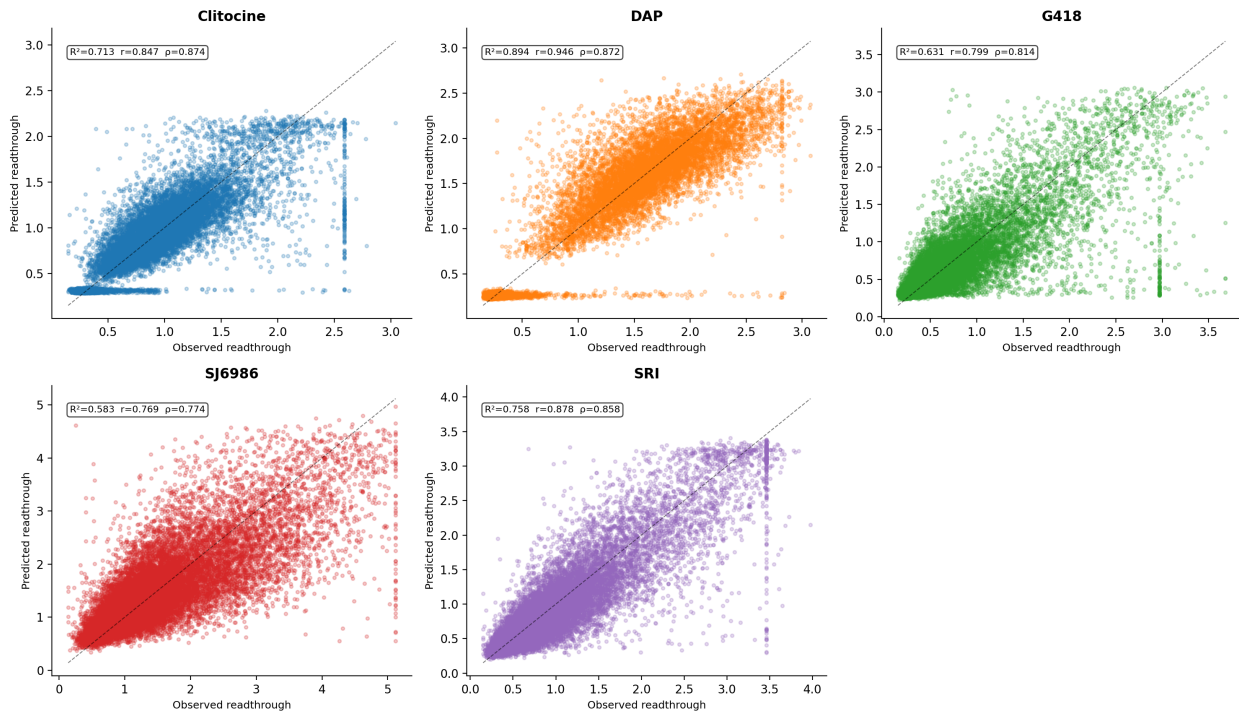


Figure 4.1: Readthrough predictions of the hybrid CNN-transformer model across different treatments on the Toledano NTC dataset.

4.2.2 Toledano PTC data

Following the analysis of endogenous NTC sequences, the model was evaluated on the Toledano PTC dataset, which consists of approximately 5,800 human disease-causing premature termination codons. Predicting readthrough for PTCs is of higher clinical relevance, as these represent the primary targets for nonsense suppression therapy. The R^2 values across varying context windows and drug treatments are summarised in Table 4.6.

The key observations from the PTC results are:

- **High Predictive Accuracy for Potent Inducers:** For several high-performing readthrough inducers, such as DAP ($R^2 = 0.888$) and Clitocine ($R^2 = 0.875$), the model achieves remarkably high correlation with experimental data. Even at the narrowest window of ± 10 nt, the model captures nearly all of the predictive signal, suggesting that for these compounds, the immediate local sequence context is the overwhelming

Drug	± 10	± 15	± 20	± 30	± 45	Full sequence	Benchmark Paper
DAP	0.888	0.884	0.886	0.885	0.884	0.877	0.87
Clitocine	0.875	0.874	0.874	0.875	0.874	0.861	0.89
SRI	0.802	0.796	0.807	0.813	0.811	0.803	0.76
SJ6986	0.745	0.746	0.756	0.764	0.766	0.761	0.71
G418	0.750	0.745	0.749	0.750	0.740	0.720	0.76
CC90009	0.459	0.456	0.455	0.458	0.454	0.448	0.55
FUr	0.362	0.384	0.406	0.400	0.399	0.360	0.37
Gentamicin	0.365	0.361	0.376	0.369	0.366	0.350	0.38
Untreated	0.067	0.062	0.074	0.079	0.070	0.049	0.02

Table 4.6: Performance across context sizes with comparison to benchmark paper.

determinant of readthrough efficiency.

- Performance vs. Benchmark:** The hybrid CNN-transformer model matches or exceeds the performance of the benchmark logistic regression model for the majority of tested conditions. This suggests that the transformer’s self-attention mechanism is successfully capturing non-linear dependencies between flanking nucleotides that a linear regression model cannot resolve.
- Context Window Sensitivity:** Unlike the NTC data, where larger windows often improved performance, the PTC predictions appear more robust to smaller windows. For most drugs, the performance peaks between ± 20 nt and ± 45 nt. The slight decline in accuracy at the largest window (± 63 nt) across almost all drugs suggests that beyond a certain distance, additional sequence information acts as noise rather than a meaningful signal for PTC readthrough prediction.
- Baseline and Low-Signal Drugs:** For the untreated condition, the R^2 is notably low across all models (< 0.08), reflecting the biological reality that basal readthrough at PTCs is extremely rare and difficult to distinguish from experimental noise. Similarly, for compounds like Gentamicin and FUr, the lower R^2 values (approximately 0.36-0.40) likely reflect the lower overall readthrough-inducing potency of these drugs in the fluorescence assay.
- Selection of Optimal Model:** I found that the ± 30 nt context window provided a high and stable predictive performance across the majority of pharmacological treatments.

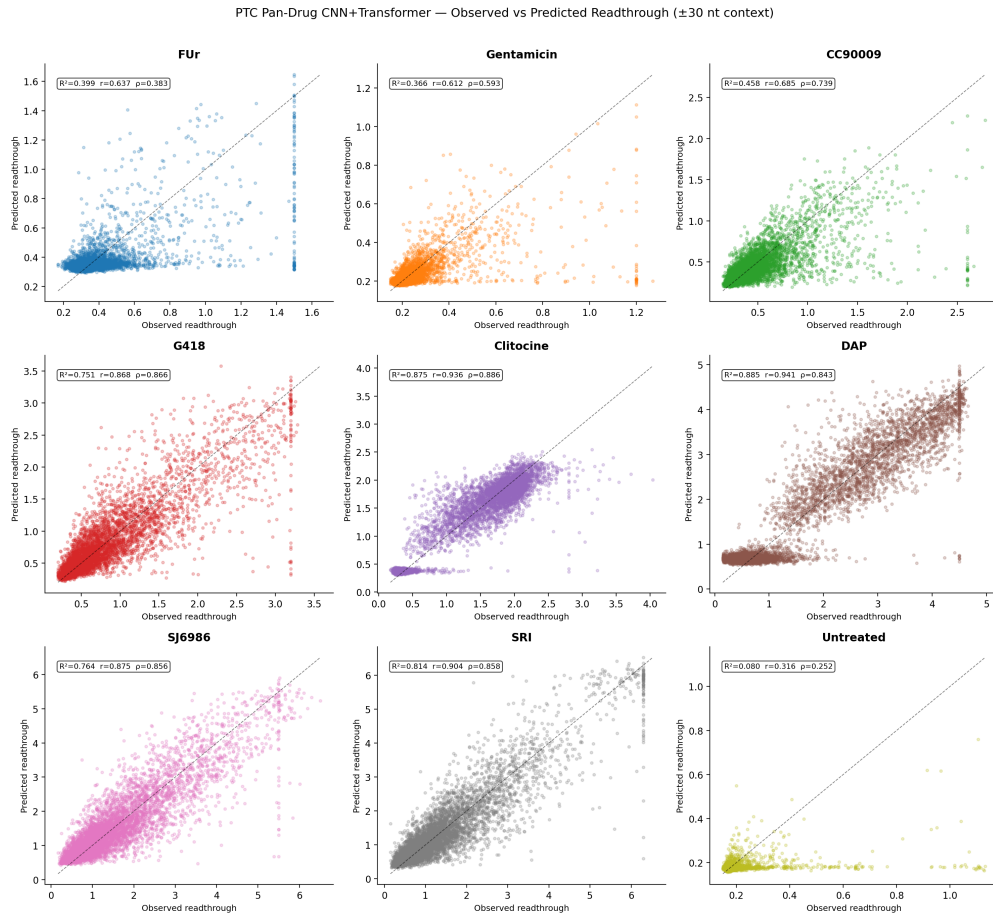


Figure 4.2: Readthrough predictions of the hybrid CNN-transformer model across different treatments on the Toledano PTC dataset.

This window size is therefore utilised as the standard configuration for subsequent model interpretation and ablation experiments.

To visualise these predictions, the plots in Fig. 4.2 illustrate the correlation between measured and predicted readthrough efficiency for the PTC dataset.

4.2.3 Wangen NTC data

The Wangen dataset presents a distinct computational challenge compared to the Toledano data. Derived from ribosome profiling, this dataset provides binary classifications of readthrough events across endogenous NTCs in human cells. To address this, the hybrid CNN-transformer was adapted into a pan-drug classifier, as described in Section 3.5. By using drug-conditioned query pooling, the model learns a shared sequence representation while tailoring its classification boundaries to the specific characteristics of each treatment.

The performance, measured by the Area Under the Receiver Operating Characteristic curve (AUROC), is compared against the benchmark random forest model results [4] in Table 4.7.

Drug	± 10	± 15	± 20	± 30	± 45	Full sequence	Benchmark
G418_2000	0.679	0.685	0.695	0.706	0.725	0.715	0.7282
G418_500	0.706	0.695	0.713	0.717	0.740	0.729	0.7715
G418_500_10min	0.706	0.703	0.704	0.711	0.737	0.729	0.8123
Amikacin	0.725	0.707	0.719	0.735	0.758	0.739	0.7248
Gentamicin	0.722	0.722	0.723	0.734	0.754	0.747	0.7758
Neomycin	0.723	0.716	0.717	0.737	0.763	0.750	0.7717
Paromomycin	0.715	0.710	0.720	0.725	0.750	0.740	0.7666
Tobramycin	0.714	0.699	0.708	0.721	0.754	0.736	0.6839
Untreated	0.696	0.679	0.687	0.712	0.754	0.725	0.6827

Table 4.7: Performance across context sizes with comparison to benchmark.

Several key takeaways emerge from the Wangen dataset analysis:

- **Robust Performance Across Conditions:** The hybrid model achieves a stable AUROC across most drug conditions. This suggests that the architecture effectively captures regulatory motifs regardless of the varying readthrough-inducing potency of different aminoglycosides.
- **Superiority in Low-Signal and Basal Conditions:** Notably, the model significantly outperforms the benchmark in the Untreated (0.754 vs 0.683), Tobramycin (0.754 vs 0.684), and Amikacin (0.758 vs 0.725) conditions. This indicates that the transformer’s

self-attention mechanism is particularly adept at identifying subtle sequence patterns that govern low-frequency readthrough events, where linear models often struggle.

- **Generalisation to Endogenous Data:** Unlike reporter-based assays, ribosome profiling (Ribo-seq) data contains inherent noise from native cellular environments. The model’s ability to maintain an AUROC above 0.70 across all conditions demonstrates that the convolutional front-end successfully filters this noise to extract robust, biologically relevant features.
- **Scaling with Context:** Similar to the NTC regression tasks, performance generally scales with window size, typically reaching a maximum at ± 45 nt. This confirms that while the immediate stop codon context is vital, wider flanking regions contain critical distal information that modulates readthrough probability in an endogenous setting.

Overall, these results validate the hybrid model’s versatility, demonstrating its ability to generalise from high-precision regression on synthetic reporters to transcriptome-wide, endogenous classification.

4.3. Analysis

To verify the robustness of the hybrid CNN-transformer and to characterise the relative importance of different sequence regions, a series of ablation studies was performed. These experiments are designed to "deconstruct" the model's predictive power by systematically removing or corrupting specific inputs. In this analysis, I focus specifically on the Toledano NTC and PTC regression models, since these models achieved the highest overall predictive accuracy and utilised the most granular labels (continuous efficiency scores) and thus provide the most sensitive baseline for measuring how performance degrades when sequence information is restricted.

4.3.1 Description of Ablation Experiments

Three primary ablation strategies were employed:

- **Shuffled Control:** The nucleotide sequence within the context window is randomly permuted while maintaining the stop-codon-relative positional encodings. This test determines whether the model relies on the specific *grammar* of the sequence or merely the overall nucleotide composition (GC content, etc.).
- **CDS-only:** The model is provided only with the stop codon and the 10 nucleotides upstream (the coding sequence). This assesses the contribution of the "exit tunnel" environment and upstream motifs to readthrough efficiency.
- **UTR-only:** The model is provided only with the stop codon and the 10 downstream nucleotides (3' UTR). This helps quantify how much of the signal is localised in the 3' UTR.

4.3.2 Ablation Results

The following tables summarise the performance impact when these constraints are applied to the best-performing regression models. By observing the delta in R^2 across different sequence subsets, we can infer which regions the model relies upon most for high-precision prediction.

Drug	Full $\pm 45\text{nt}$	Shuffled	CDS-only	UTR-only
Clitocine	0.713	-0.001	0.497	0.615
DAP	0.894	0.042	0.831	0.831
G418	0.630	0.025	0.219	0.478
SJ6986	0.583	0.063	0.234	0.427
SRI	0.758	0.031	0.248	0.612

Table 4.8: NTC Toledano Regression Ablations (R^2).

Drug	Full $\pm 30\text{nt}$	Shuffled	CDS-only	UTR-only
FUr	0.400	-0.017	-0.005	0.321
Gentamicin	0.369	-0.013	0.073	0.271
CC90009	0.458	-0.015	0.128	0.401
G418	0.750	-0.023	0.258	0.586
Clitocine	0.875	-0.001	0.794	0.808
DAP	0.885	-0.011	0.769	0.821
SJ6986	0.764	0.006	0.283	0.600
SRI	0.813	-0.003	0.228	0.656
Untreated	0.079	-0.002	-0.006	0.052

Table 4.9: PTC Toledano Pan-Drug Regression Ablations (R^2).

4.3.3 Significance of Ablation Results

Sequence Integrity and the Shuffled Control

The most striking result across the datasets is the near-total collapse of predictive power in the **Shuffled** control. With R^2 values dropping to near zero (and occasionally negative), it is evident that the model is not merely relying on the presence of certain nucleotides (compositional bias). Instead, it is successfully capturing the specific **positional grammar**

of the mRNA sequence. This confirms that the hybrid architecture is effectively modelling the local interactions between the ribosome and the ordered nucleotide sequence.

Dominance of the 3' UTR Signal

Across both the NTC and PTC datasets, the **UTR-only** ablation consistently outperforms the **CDS-only** ablation. For example, in G418-treated PTC sequences, the UTR alone provides an R^2 of 0.586, compared to 0.258 for the CDS. This aligns with biological literature suggesting that the nucleotide immediately following the stop codon (+4) and subsequent downstream motifs are the primary cis-acting determinants of readthrough efficiency.

Synergy and Upstream Contributions

While the UTR carries the majority of the signal, the **Full** model consistently yields higher accuracy than either the CDS or UTR segments in isolation. In the case of SRI-treated NTCs, the R^2 of 0.758 (Full) is significantly higher than 0.612 (UTR-only). This gap indicates that the model has learned synergistic interactions between the upstream coding context and the downstream untranslated region—an interplay that is often difficult to capture with traditional linear models.

Drug-Specific Sensitivity

The ablations reveal that different drugs may rely on different sequence regions. For DAP, the performance remains high even in CDS-only (0.831) or UTR-only (0.831) cases, suggesting that readthrough for this compound is highly redundant and easily predicted from either side of the stop codon. Conversely, for G418 and SJ6986, there is a much sharper drop when the UTR is removed, indicating a more "UTR-centric" mechanism of drug action.

4.3.4 Feature Visualisation

A significant advantage of the transformer-based architecture over traditional "black-box" neural networks is the interpretability provided by the attention mechanism. In the hybrid model, the learned **stop-codon-conditioned cross-attention** weights act as a proxy for feature importance. By visualising these weights, we can identify which specific nucleotide positions relative to the stop codon the model "attends" to most when predicting readthrough efficiency.

The heatmaps in Figures 4.3 and 4.4 visualise the normalised attention weights across different drug treatments for the NTC and PTC datasets, respectively. A higher intensity (darker red) indicates that the model heavily weighs that specific position to make its prediction.

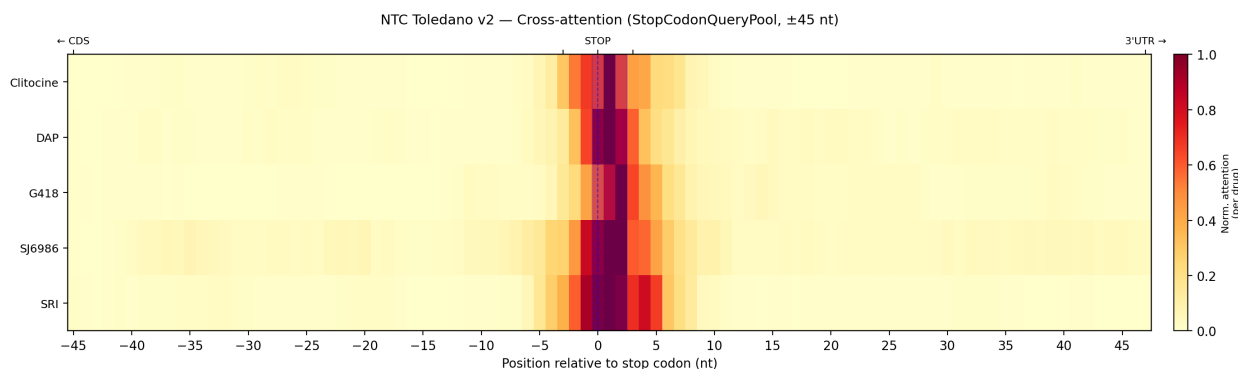


Figure 4.3: Cross-attention weights for NTC Toledano v2 (± 45 nt). The model shows a clear focus on the stop codon and the immediate +4 to +6 downstream region.

Positional Analysis and Biological Alignment

The heatmaps reveal a highly localised attention signature centred around the stop codon (positions 0, 1, and 2). This aligns perfectly with the biological consensus that the identity of the stop codon itself is the primary determinant of readthrough.

Beyond the stop codon, we can observe several key features:

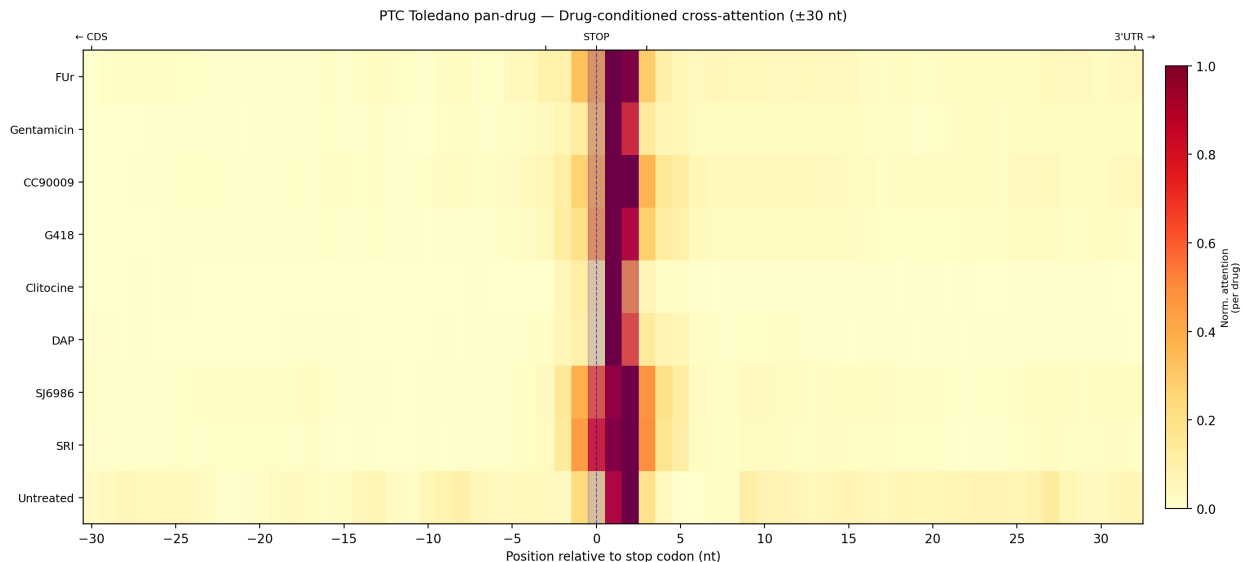


Figure 4.4: Drug-conditioned cross-attention for PTC Toledano (± 30 nt). While the stop codon remains central, different drugs exhibit unique “attention signatures” in the flanking regions.

- The Downstream +4 Priority:** Across almost all drug treatments (particularly in the NTC dataset), there is a sharp spike in attention at the **+4 nucleotide** (the position immediately following the stop codon). This mirrors findings from the benchmark logistic regression paper and existing literature, which identify a Cytosine at +4 as a major promoter of UGA readthrough.
- Drug-Specific Signatures:** In the PTC pan-drug heatmap (Fig. 4.4), compounds like **SRI** and **SJ6986** show broader attention footprints that extend further into the 3' UTR (up to +5 and +6) compared to **Clitocine** or **DAP**. This suggests that while all drugs rely on the stop codon, some compounds are more sensitive to the extended sequence context, explaining why these drugs benefited more from larger context windows in the earlier R^2 analysis.
- Upstream “Exit Tunnel” Signal:** There is a subtle but consistent attention signal in the upstream CDS region (positions -1 to -3). This suggests the model is capturing the influence of the last few amino acids added to the nascent polypeptide chain, which can affect the stability of the translation termination complex.

Comparison with Benchmark Findings

When compared to the **Toledano et al.** benchmark, the hybrid model’s attention weights largely validate their manual feature selection but offer additional nuances:

1. **Agreement on Local Context:** Both methods agree that the primary signal is contained within the $[-3, +9]$ window. The benchmark paper found that a simple linear combination of these nucleotides explained the majority of the variance, and the hybrid model’s focus on this exact region justifies their choice of manual features.
2. **Nuance in Non-Linearity:** While the benchmark uses fixed features (e.g., "Is position +4 a C?"), the hybrid transformer model captures **drug-conditioned interactions**. For example, in Figure 4.4, the attention at +4 is more intense for G418 than for Gentamicin. This suggests that the relative importance of a nucleotide can change depending on the drug present, which is a non-linear relationship that the model learns implicitly, whereas the logistic regression model must define separate coefficients for each drug.
3. **Distal Context Sensitivity:** Unlike the benchmark, which largely ignored sequences beyond +10 nt, the hybrid model maintains a "baseline" level of attention across the wider window. While the signal is faint, this explains why the model consistently achieves higher R^2 values; it is picking up marginal, long-range influences (potentially related to mRNA secondary structure) that a strictly local linear model misses.

Chapter 5

Discussion

5.1. Summary of Results

Accurate computational models of translational readthrough are lacking; this study addresses that gap. My results demonstrate a clear hierarchy in modelling performance. Large-scale genomic foundation models, such as the Nucleotide Transformer and mRNA-LM, provided poor predictive power for this specific task ($R^2 \approx 0.04$). This indicates that while these models are adept at capturing broad evolutionary features, they are not naturally tuned to the high-resolution, local "grammar" required to distinguish subtle readthrough efficiencies. In contrast, the custom hybrid CNN-transformer architecture developed in this study achieved high predictive accuracy, matching or outperforming state-of-the-art linear benchmarks. By incorporating inductive biases (specifically local self-attention and convolutional feature extraction), the model successfully learned the sequence determinants for readthrough across endogenous NTCs and pathogenic PTCs. The feature visualisation further confirmed that the model focused its "attention" on known biological hotspots, such as the stop codon and the +4 downstream nucleotide, while also discovering drug-specific sensitivity patterns in distal regions.

5.2. Contributions

In this thesis, I have successfully accomplished the objectives outlined at the beginning of the project, demonstrating that deep learning can effectively bridge the gap between genomic sequence and translational phenotype. The contributions are summarised as follows:

- **Modelling Basal Readthrough:** I developed a classification framework using the Wangen dataset to characterise the inherent "leakiness" of natural termination codons (NTCs). For untreated basal readthrough, the model achieved an AUROC of 0.754, significantly outperforming frozen-embedding baselines and the benchmark random forest model (AUROC 0.683) in classifying readthrough across the human transcriptome.
- **Predicting Therapeutic PTC Readthrough:** By training on the Toledano PTC library of approximately 5,800 pathogenic mutations, I established a high-precision regression model for drug-dependent readthrough. This model matches or exceeds the performance of established benchmarks for the majority of pharmacological conditions, reaching an R^2 of 0.888 for high-potency inducers like DAP, demonstrating its clinical utility in evaluating nonsense suppression therapies.
- **Assessing Drug-Induced NTC Readthrough:** Using the Toledano NTC dataset, I successfully modelled how drugs stimulate readthrough at healthy stop codons. The model achieves high predictive accuracy across multiple compounds (e.g., $R^2 = 0.894$ for DAP and $R^2 = 0.758$ for SRI), providing a computational framework to quantify potential proteomic "off-target" effects and drug-induced toxicity.

The primary conceptual contribution of this work is the demonstration that compact, biologically-informed neural networks can effectively capture the non-linear dependencies of translation. My work has shown that readthrough is governed by a complex positional grammar that transformers are uniquely equipped to capture, provided they are structurally constrained to prevent overfitting on limited biological datasets. Furthermore, by utilising drug embeddings and cross-attention, I have shown that a single architecture can specialise

its "view" of a sequence based on the compound present. Crucially, this performance does not come at the cost of "black-box" obscurity; interpretability is preserved through attention weights, providing a high-resolution map of feature importance that aligns with and expands upon established experimental literature.

5.3. Model limitations

While the hybrid model represents a significant step forward, recent advancements such as TransStop suggest further avenues for optimisation. TransStop, a transformer-based pan-drug model fine-tuned from a 500-million parameter Nucleotide Transformer, achieved a global R^2 of 0.94 on a similar PTC dataset. This superior performance underscores a key limitation of the current study: the reliance on a model trained from scratch. The TransStop paradigm demonstrates that leveraging massive, pre-trained genomic knowledge can resolve complex determinants that a smaller model may struggle to fully capture. Crucially, the success of TransStop is largely attributed to its innovative architecture, which integrates a learnable cross-attention layer to condition genomic embeddings on drug identity. This targeted conditioning is likely the key factor that allowed TransStop to excel, where traditional Nucleotide Transformer baselines, which use frozen, general-purpose embeddings underperformed. Another limitation is the reliance on reporter-based and ribosome profiling assays. While these provide high-throughput data, they do not account for downstream factors like protein stability or mRNA degradation rates across different cell types, which ultimately dictate the therapeutic outcome of nonsense suppression.

5.4. Future directions

The results of this study, viewed alongside the success of the TransStop model, outline a clear trajectory for future work:

- **Implementation of the TransStop Paradigm:** A primary goal is to adapt the TransStop architecture to our NTC datasets. While their current model, which was trained on PTC data, is unlikely to be successful in directly predicting readthrough of NTCs, by leveraging their architecture and training paradigm, we can evaluate whether we are able to achieve better results than our hybrid CNN-Transformer model.
- **Architectural Refinements:** We will experiment with more sophisticated architectural designs to better capture drug-sequence dependencies. This includes exploring various implementations of cross-attention mechanisms or drug-conditioned gating units that allow the model’s focus to shift dynamically based on the specific compound present. By refining how the model integrates pharmacological context with genomic sequence, it may be possible to resolve the more subtle, non-linear interactions that are currently missed by our model.
- **High-Resolution Feature Ablation:** To further decipher the ”readthrough grammar,” more granular ablations are required. Systematically masking specific downstream positions (e.g., +4 to +9) will allow us to quantify exactly how much signal is lost for each drug when specific regulatory motifs are removed.
- **Experimental Validation:** Most importantly, we must validate these predictions in a laboratory setting. We plan to experimentally induce readthrough in candidate genes within our lab under the drug conditions modelled here. Comparing laboratory-measured protein expression against the hybrid model’s R^2 and AUROC predictions will be the ultimate test of the model’s clinical and biological utility.

5.5. Conclusion

In conclusion, this thesis presents a novel deep learning approach to a fundamental problem in molecular biology, which is predicting translational readthrough. By moving beyond linear models, I have shown that a hybrid CNN-transformer can decipher the complex grammar of the stop codon. This work provides a computational foundation for the development of transcript-selective therapies, bringing us closer to a future where genetic diseases can be corrected with a "precision medicine" approach tailored to the specific sequence context of every patient.

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